



Semester 2 Examinations 2014/2015

Exam Code(s)	4BME1, 3BMS2, 4BMS2, 3BS9, 4BS2/1HA1.
Exam(s)	Third & Fourth Science and postgraduate.
Module(s)	Numerical Analysis II.
Module Code(s)	MA378, MA531
Paper No	1.
Repeat Paper	No
External Examiner(s)	Dr. M. Lawson.
Internal Examiner(s)	Prof. G. Ellis, Dr. M. Hayes.

Instructions:

Answer 3 questions.
All questions carry equal marks.

Duration	2 hours.
No. of Pages	5 pages.
Department(s)	Mathematics.
Course Co-ordinators(s)	

Requirements:

Release in Exam Venue	Yes ☒	No ☐
MCQ	Yes ☐	No ☒

Handout	
Statistical Tables/ Log Tables	Yes.
Cambridge Tables	
Graph paper	
Log Graph Paper	
Other Materials	

Students may use their own electronic calculators which must not be capable of storing text.

1. (a) (i) Show that there is at most one polynomial of degree $\leq n$ that solves the Polynomial Interpolation Problem. (Explain all your steps in full.)

- (ii) Given $n + 1$ distinct points ($n \geq 2$) with

$$x_0 = 0, x_1 = 1, \dots, x_n = n;$$

and defining

$$L_i = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j},$$

hence, or otherwise, show

$$L_1(x) + 2^2 L_2(x) + 3^2 L_3(x) + \dots + n^2 L_n(x) = x^2.$$

(Explain all steps in full.)

part (a) consists of 40% of Q.1

- (b) (i) Write down the Lagrange form of the interpolation polynomial to

$$f(x) = e^x,$$

at interpolating points $\{-1, 0, 1\}$.

- (ii) Find the error in Q1(b)(i) at $x = 0.5$.

- (iii) Cauchy's theorem is:

$$f(x) - p_n(x) = \frac{f^{n+1}(\tau)}{(n+1)!} \Pi_{n+1}(x),$$

where τ has the usual meaning, and

$$\Pi_{n+1}(x) = \prod_{i=0}^n (x - x_i).$$

Use a form of Cauchy's theorem to find an upper bound on the error over the entire domain in Q1(b)(i).

part (b) consists of 40% of Q.1

- (c) Prove Cauchy's theorem (which is stated in Q1(b)(iii)). You may find the following function of use:

$$\phi(t) = f(t) - p_n(t) - \frac{f(x) - p_n(x)}{\Pi_{n+1}(x)} \Pi_{n+1}(t).$$

part (c) consists of 20% of Q.1

2. (a) Define the *linear spline interpolant* which interpolates a function $f(x)$ at the points $a = x_0 < x_1 < \dots < x_n = b$.

part (a) consists of 10% of Q.2

- (b) Suppose that $l(x)$ is the linear spline interpolant as in (a) and that the points are evenly spaced, with $h = x_i - x_{i-1}$. Use Cauchy's theorem (which is stated in Q1(b)(iii)) to give an upper bound for $\|f - l\|_\infty$ in terms of h and $\|f''\|_\infty$.

part (b) consists of 30% of Q.2

- (c) Let $f(x) = \sin(x)$, $a = 0$ and $b = \frac{\pi}{2}$. What value of n would you need to ensure that $\|f - l_n\|_\infty < 10^{-6}$, where l_n is the linear spline interpolant which interpolates the function $f(x)$ over the interval $[a, b]$.

part (c) consists of 20% of Q.2

- (d) Let $f \in C^1[a, b]$ and let $a = x_0 < x_1 < \dots < x_n = b$ for $n \geq 2$. Define the Hermite cubic spline $s(x)$, interpolating $f(x)$ at the knot points $x_0, \dots, x_n$, which are assumed to be evenly spaced with $h = x_i - x_{i-1}$.

Show that if the Hermite cubic spline is given on the interval $[x_{i-1}, x_i]$ by

$$s(x) = c_0 + c_1(x - x_{i-1}) + c_2(x - x_{i-1})^2 + c_3(x - x_{i-1})^3,$$

then

$$\begin{aligned} c_0 &= f(x_{i-1}), \quad c_1 = f'(x_{i-1}), \\ c_2 &= \frac{3(f(x_i) - f(x_{i-1}))}{h^2} - \frac{f'(x_i) + 2f'(x_{i-1}))}{h}, \\ c_3 &= \frac{f'(x_i) - f'(x_{i-1}))}{h^2} - \frac{2(f(x_i) - f(x_{i-1}))}{h^3}. \end{aligned}$$

part (d) consists of 40% of Q.2

3. (a) Explain Newton-Cotes quadrature, and show how to derive the 2-point and 3-point Newton-Cotes quadrature rules (i.e. the trapezium method and Simpson's method) for approximating the integral

$$\int_{-1}^1 f(x)dx.$$

part (a) consists of 30% of Q.3

- (b) Use the composite Trapezoid rule with 4 subintervals to approximate

$$\int_0^1 \sin(x)dx.$$

Give the exact absolute value of the error and compute the upper bound on the absolute value of the error using the following formula:

$$\frac{(b-a)^3}{n^2}M_2,$$

where the symbols have the usual meaning.

part (b) consists of 35% of Q.3

- (c) Given the Hermite polynomial

$$p_{2n+1}(x) = \sum_{i=0}^n f(x_i)H_i(x) + \sum_{i=0}^n f'(x_i)K_i(x),$$

where

$$H_i(x) = [L_i(x)]^2[1 - 2L'_i(x_i)(x - x_i)],$$

and

$$K_i(x) = [L_i(x)]^2(x - x_i).$$

Show the Gauss quadrature:

$$\int_a^b w(x)f(x)dx \approx \sum_{i=0}^n W_i f(x_i),$$

where

$$W_i = \int_a^b w(x)[L_i(x)]^2 dx,$$

and $w(x)$ and $f(x)$ have the required properties. Explain all your steps clearly.

part (c) consists of 35% of Q.3

4. (a) Consider the following boundary value problem

$$-u''(x) + 3u(x) = x, \quad u(0) = u(3) = 0.$$

Use the finite element method with piecewise linear basis functions on the mesh $\{0, 1, 2, 3\}$ to find an approximation of the solution to this problem.

part (a) consists of 80% of Q.4

- (b) Given that $p \in C^1[a, b]$, $r \in C[a, b]$, $f \in L^2(a, b)$ and $p(x) \geq c_0 > 0$, $r(x) \geq 0$, $\forall x \in [a, b]$.

Show that $u \in H_E^1(a, b)$ is a weak solution to

$$-\frac{d}{dx} \left(p(x) \frac{du}{dx} \right) + r(x)u = f(x), \quad a < x < b,$$

with boundary conditions: $u(a) = A$, $u(b) = B$ if u satisfies the Galerkin principle, that is:

$$\int_a^b p(x) \frac{du}{dx} \frac{dv}{dx} dx + \int_a^b r(x)uv dx = \int_a^b f(x)v dx,$$

$\forall v \in H_0^1(a, b)$.

part (b) consists of 20% of Q.4